

The Hall Of Residence Application/Reservation Programme

Description of Programme: The Hall of Residence Application/Reservation Programme is a Java application used to manage the process of issuing rooms to students registering for hall residency and store information of residents. The programme is executed by compiling and running Application.java or by launching the Hall Residency.jar . After successful execution, the user will first be presented with a welcome frame, which allows the user to select a button if they wish to apply for residency as a new or returning student, reserve a room as a returning student or be able to edit, delete and sort residents' information as an administrative staff member. If the user selects the "Apply" button, they will be required to fill out an application form. If there are available rooms, the user will be assigned a room based on their three hall choices in order of 1st, 2nd and 3rd, budget and gender. If the user is a male, he will be assigned a random room number between 1-20 inclusive, as the first 20 rooms on each of the four blocks (A, B, C or D) are dedicated to males. However, if the user is a female, she will be assigned a random room number between 21-50 inclusive, as the remaining 30 rooms on each of the four blocks are dedicated to females. After being assigned a room, a confirmation message will appear. If the user does not have enough funds or there are no available rooms, a message stating that the user cannot be assigned a room. If the user selects the "Reserve" button, they will be required to fill out a reservation form. If their preferred hall and block have available rooms and they have enough funds, they will be assigned a room based on their gender using the same procedure explained prior. After being assigned a room, a confirmation message will appear. If the user does not have enough funds or there are no available rooms in their preferred hall, a message stating that the user cannot be assigned a room. If the user selects the "Admin Login" button, they will be required to enter their name, identification number and password, which will be cross referenced with the set names, identification numbers and passwords that have been granted access. If the inputs are the same as the any of the set values, the user will gain access to a table displaying the residents' information. The user will be able to sort the information by first name, hall and faculty. They will be also be able to edit and delete residents' information.

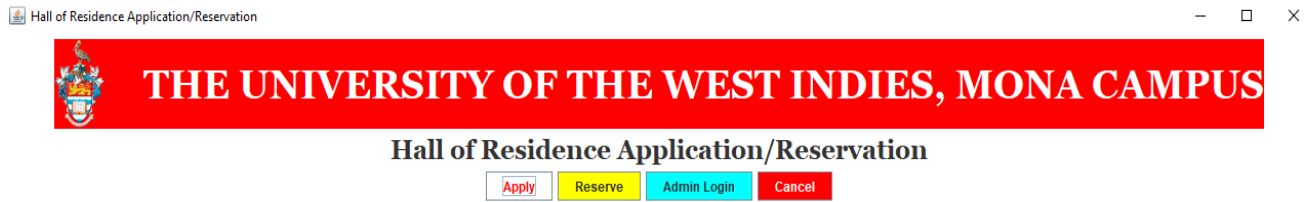


Figure 1. *Initial screen presented to the user*

Figure 1 presents the first frame, which displays a welcome screen titled "Hall of Residence Application/Reservation." This frame gives:

- new and returning students the option to apply for a room by selecting the "Apply" button,
- Returning students the option to reserve their past room for the upcoming year by clicking on the "Reserve" button,
- staff members have the option to view the list of students who have registered for rooms by clicking the "Admin Login" button then signing in, and the "Cancel" button to close the frame.

Figure 2A. *The second screen the user will be presented with if they wish to apply for a room*

Figure 2B. *A continuation of Figure 2A*

Figures 2A and 2B show the application form for new and returning students. This frame holds the form to be filled out if the user is applying for a room. After filling out the form, the user can either select the “Submit” button which will display a frame that either confirms that the user’s application was successful and shows the hall, block and room number that the user is assigned or declines the user’s application if either the halls are fully occupied or the user does not have enough funds. The user can also select the “Cancel” button, which closes the frame.

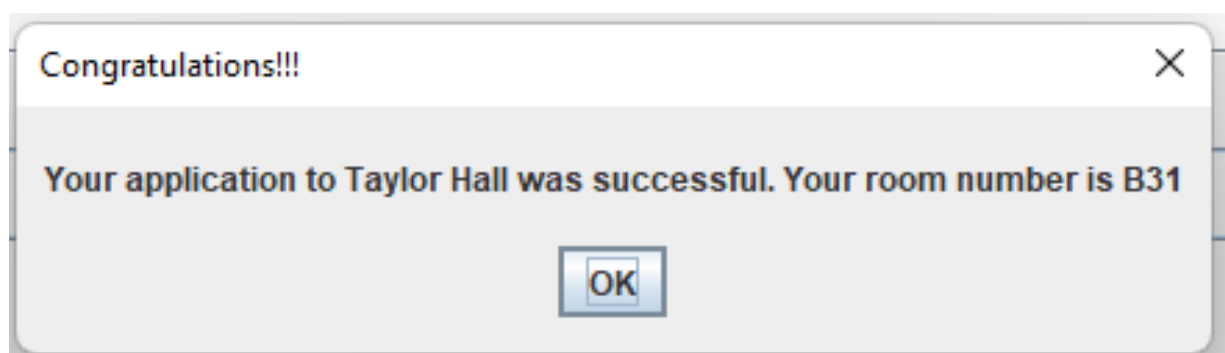


Figure 3. *The third screen that the user will see if their application was successful.*

After the successful submission of the application form, students will be presented with a message indicating further information about their assigned hall and room number. The room

number is given in the format (Block+RoomNumber). The user can then select the “OK” button, which closes the frame.

The screenshot shows a web form titled "Hall of Residence Reservation Form for Returning Students ONLY". The form includes the following fields and options:

- Name: [Text input]
- ID: [Text input]
- Age: [Text input]
- Sex: ☐ Male ☐ Female
- Telephone: (example: 876-123-4567) [Text input]
- Email: [Text input]
- Are you an undergraduate or postgraduate student? ☐ Undergraduate ☐ Postgraduate
- Choose your Faculty: Engineering [Dropdown menu]
- Major? [Text input]
- Select your upcoming year: ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8
- Select your Hall of Choice: Elsa Leo Rhynie Hall [Dropdown menu]
- Please state your Block preference: A [Dropdown menu]
- Budget: (Rex: \$100000; Taylor: \$150000; Elsa: \$200000) [Text input]

At the bottom of the form are two buttons: "Cancel" (red) and "Submit" (green).

Figure 4: *The second screen the user will be presented with if they wish to reserve a room*

Figure 4 shows the reservation form that is to be filled out by returning students if they would like the room they previously held. After filling out the form, the user can either select the “Submit” button, which will display a frame that either confirms that the user’s reservation was successful and shows the hall, block and room number that the user is assigned or declines the user’s application if either their preferred hall is fully occupied or the user does not have enough funds. The user can also select the “Cancel” button, which closes the frame.

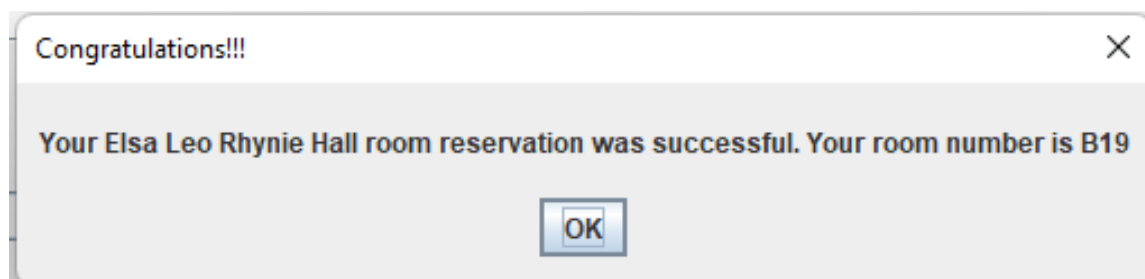


Figure 5. *The third screen that the user will see if their reservation was successful.*

After the successful submission of the reservation form, students will be presented with a message indicating further information about their assigned hall and room number. The room

number is given in the format (Block+RoomNumber). The user can then select the “OK” button, which closes the frame.

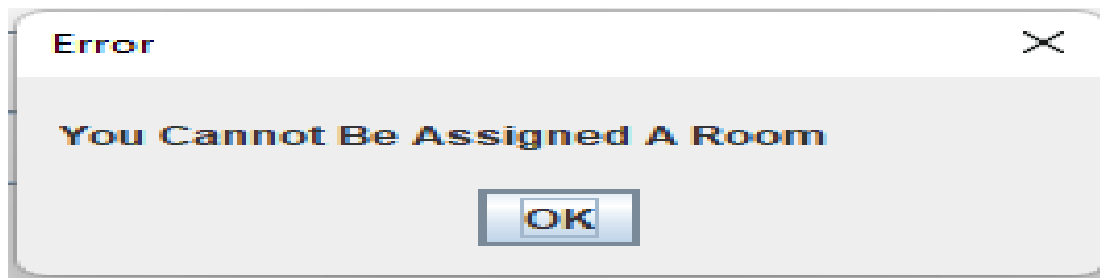


Figure 6. *The third screen that the user will see if their application/reservation was unsuccessful.*

After the submission of the reservation form, if there are no available rooms or they do not possess enough funds, students will be presented with a message indicating that their application/reservation was unsuccessful. The message will state that the user cannot be assigned a room. The user can then select the “OK” button, which closes the frame.

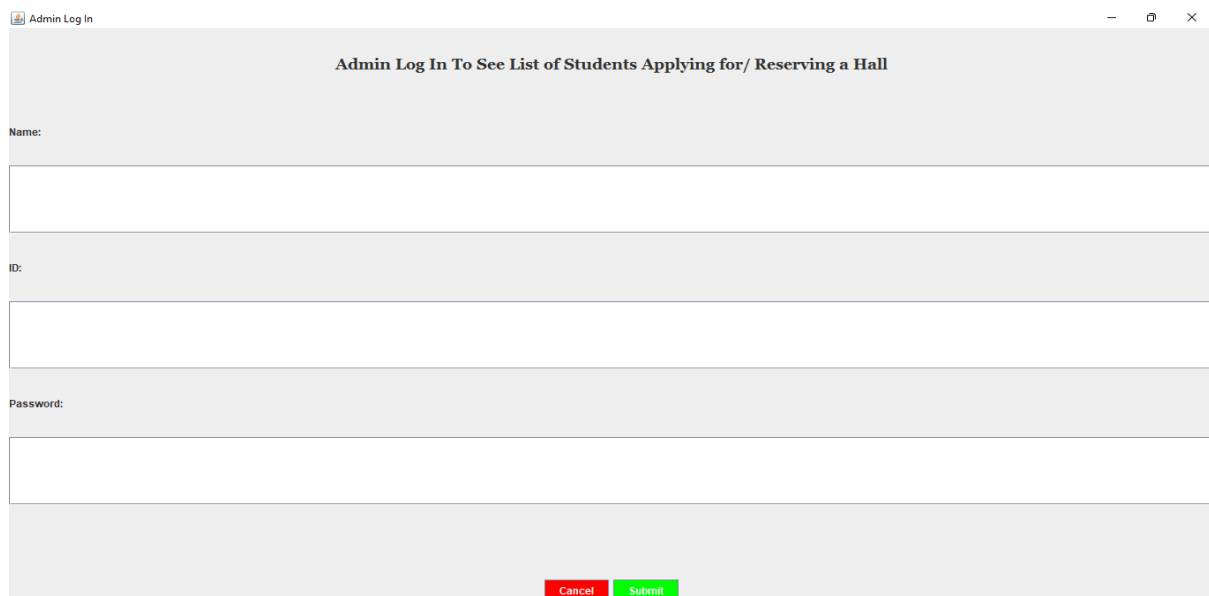
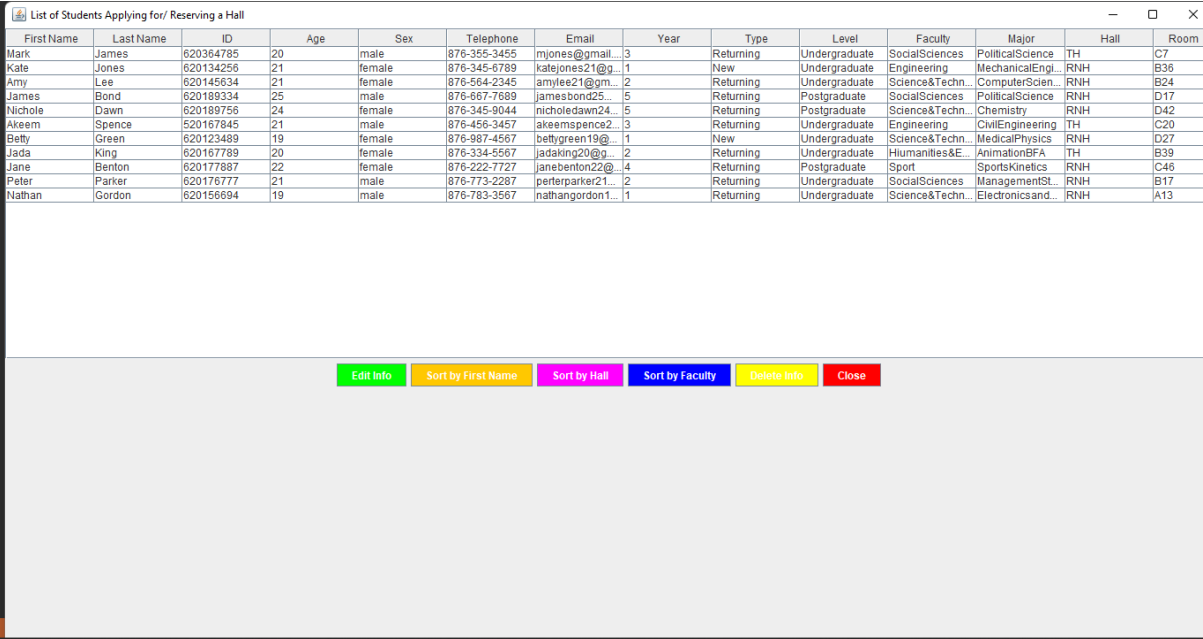


Figure 7. *The second screen the user will be presented with if they are administrators.*

The “Admin Login” button will lead the user to the frame in Figure 4, which displays the login screen for administrators. The user will input their name, work identification number and their password in their respective textboxes. If the inputs are accepted after selecting the

“Submit” button, a frame holding the list of residents will be prompted. The user can also select the “Cancel” button, which closes the frame.



First Name	Last Name	ID	Age	Sex	Telephone	Email	Year	Type	Level	Faculty	Major	Hall	Room
Mark	James	620364785	20	male	876-355-3455	mjones@gmail...	3	Returning	Undergraduate	SocialSciences	PoliticalScience	TH	C7
Kate	Jones	620134256	21	female	876-345-6789	katejones21@g...	1	New	Undergraduate	Engineering	MechanicalEngi...	RNH	B36
Amy	Lee	620145634	21	female	876-564-2345	amylee21@gm...	2	Returning	Undergraduate	Science&Techn...	ComputerScien...	RNH	B24
James	Bond	620189334	25	male	876-667-7689	jamesbond25...	5	Returning	Postgraduate	SocialSciences	PoliticalScience	RNH	D17
Nichole	Dawn	620189756	24	female	876-345-9044	nicholedawn24...	5	Returning	Postgraduate	Science&Techn...	Chemistry	RNH	D42
Akeem	Spence	520167845	21	male	876-456-3457	akeemspence2...	3	Returning	Undergraduate	Engineering	CivilEngineering	TH	C20
Betty	Green	620123489	19	female	876-987-4567	bettygreen19@...	1	New	Undergraduate	Science&Techn...	MedicalPhysics	RNH	D27
Jada	King	620167789	20	female	876-334-5567	jadaking20@g...	2	Returning	Undergraduate	Humanities&E...	Animation&FA	TH	B39
Jane	Benton	620177887	22	female	876-222-7727	janebenton22@...	4	Returning	Postgraduate	Sport	SportsKinetics	RNH	C46
Peter	Parker	620176777	21	male	876-773-2287	perterparker1...	2	Returning	Undergraduate	SocialSciences	ManagementSt...	RNH	B17
Nathan	Gordon	620156694	19	male	876-783-3567	nathangordon1...	1	Returning	Undergraduate	Science&Techn...	Electronicsand...	RNH	A13

Figure 8. The third screen the user will see if their login credentials were accepted.

This screen presents a table that holds the residents’ records, in which staff members are able to sort the table by virtue of their first name, hall and faculty. In addition, they will be offered the option to edit or delete a record. The user can also select the “Cancel” button, which closes the frame.

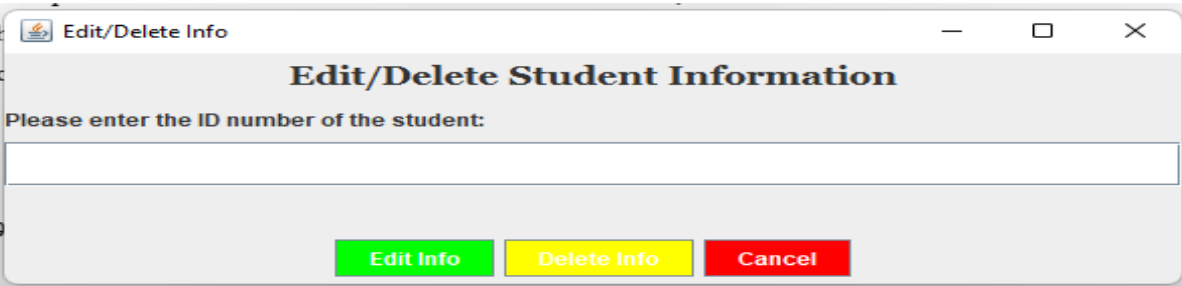


Figure 9. The fourth screen that the user will see if either the “Edit Info” button or the “Delete Info” button is selected.

This screen allows the user to enter the identification number of the student whose record they would like to edit or delete. The user can also select the “Cancel” button, which closes the frame.

Edit Info

Hall of Residence Update Local Info

Name:

ID:

Age:

Telephone: (example: 876-123-4567)

Email:

Are you a new or returning student?

☐ New ☐ Returning

Are you an undergraduate or postgraduate student?

☐ Undergraduate ☐ Postgraduate

Choose your Faculty:

Engineering

Major:

Select your year:

2

Cancel Submit Show desktop

Figure 10. *The fifth screen that the user will see if they wish to update a resident's record*

Figure 10 is prompted when the user decides to update a resident's record after entering that student's identification number. After selecting the "Submit" button, the student's information will be updated in the table. The user can also select the "Cancel" button, which closes the frame.

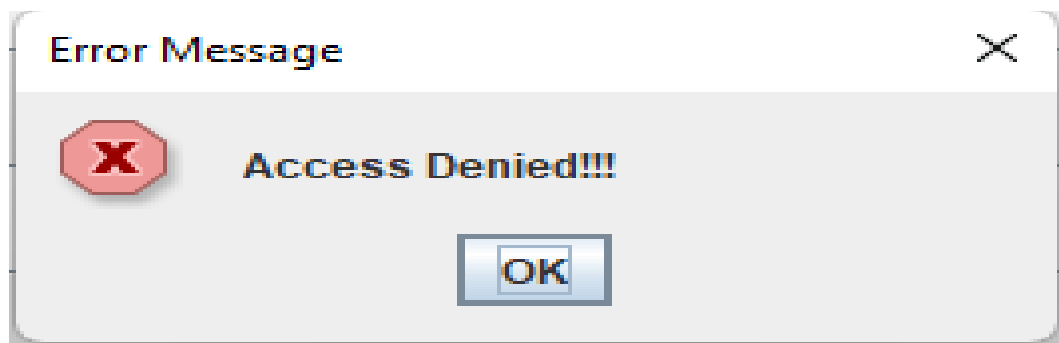


Figure 11. *The third screen the user will see if their login credentials were not accepted.*

This screen presents an error message indicating that the user's request for access to the table holding student's information has been denied due to the inputs not being accepted. The user can select the "OK" button, which closes the frame.

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